Dynamical Systems

Part II

Recorded from the lecturer in 2026-2027.

Dynamical systems – mathematical entities described by a set of equations for a set of variables w.r.t. a time-like variable.

State space or phase space.

∗ Continuous time ⇒ set of ODEs

Lotka–Volterra for rabbits (r) & sheep (s)

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad a, b, \ldots, f > 0.$$

$$\text{State space} \qquad [0, \infty) \times [0, \infty).$$

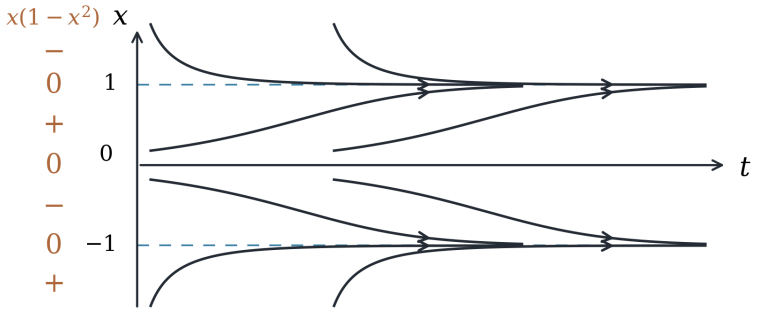
e.g. Logistic map $\qquad x_{n+1} = \mu x_n (1 - x_n), \quad 0 \le \mu \le 1$

Questions to address

- ∗ Long-term behavior of the system?
- ∗ Are there periodic solutions
- ∗ Stability of solutions, chaos
- ∗ Effect of system parameter variation

Example 1D system $\qquad \dot{x} = x(1 - x^2)$

$$\text{Solution} \qquad x(t) = \pm \left(1 + 2e^{2(t_0 - t)}\right)^{-1/2}$$



as $t \to \infty$

$x(0) > 0 \Rightarrow x(t) \to 1$

$x(0) < 0 \Rightarrow x(t) \to -1$

$x(0) = 0 \Rightarrow x(t) = 0$

Example 2D, Lotka–Volterra

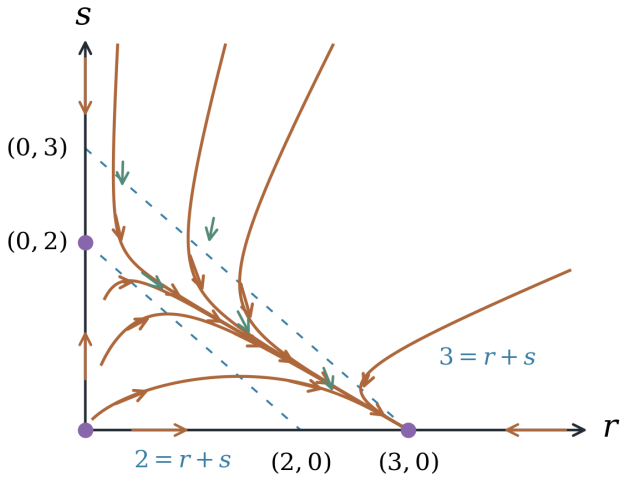
$$\dot{r} = r(3 - r - s)$$
$$\dot{s} = s(2 - r - s)$$

Can get $r(s) \Rightarrow t(s)$

∗ Find fixed points

$$\dot{r}, \dot{s} = 0 : \quad (0,0), (3,0), (0,2)$$

∗ Look at $\dot{r}, \dot{s}$ in state space



$$\text{As } t \to \infty \qquad (r,s) \to \begin{cases} (3,0), & r > 0, \\ (0,2), & r = 0,\ s > 0, \\ (0,0), & r = s = 0. \end{cases}$$

Example predator-prey

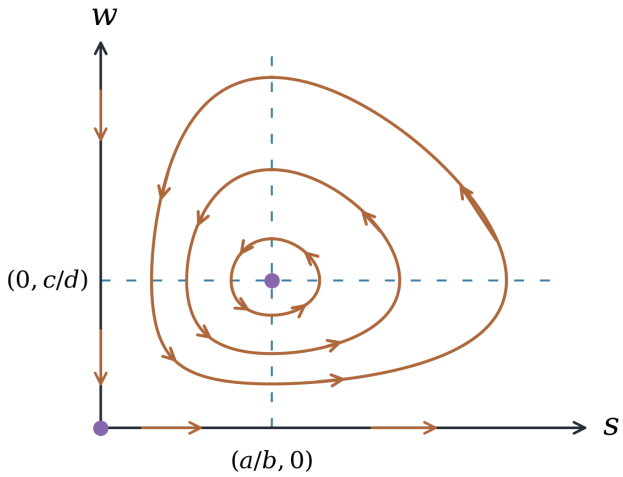
sheep (s) & wolves (w)

$$\dot{w} = w(-a + bs)$$
$$\dot{s} = s(c - dw)$$

$$a, \ldots, d > 0$$

Fixed points

$$(0,0), \quad (a/b, c/d)$$



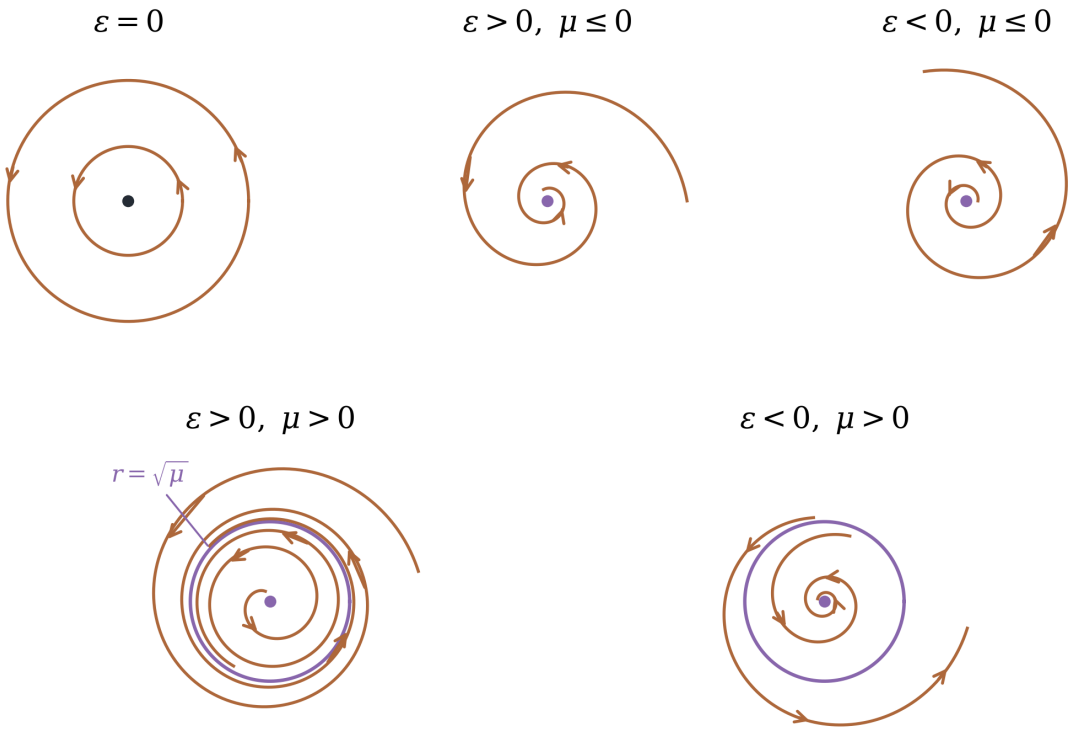
$$(s,w) \to \begin{cases} \text{periodic around } (a/b, c/d), & s > 0,\ w > 0, \\ (0,0), & s = 0,\ w \geq 0, \\ (\infty, 0), & s > 0,\ w = 0 \end{cases} \qquad \text{as } t \to \infty$$

Example

$$\dot{x} = -y + \varepsilon x(\mu - x^2 - y^2)$$
$$\dot{y} = \quad x + \varepsilon y(\mu - x^2 - y^2)$$

Switch to polars:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan} \frac{y}{x},$$
$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \qquad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} \quad \Rightarrow$$
$$\underline{\dot{r} = \varepsilon r(\mu - r^2)}, \qquad \underline{\dot{\theta} = 1}$$



End of Lecture 1